Research Statement

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Human Computer Interaction, Participatory Design, Collaborative Technologies, and UX Research

Design research plays a crucial role in ensuring that emerging technologies are shaped not only by technical innovation but also by the needs, experiences, and values of the people who use them. Through participatory and human-centered approaches, design research enables communities to actively contribute to the development of technologies that influence their daily lives, leading to solutions that are more inclusive, usable, and socially responsible. As design research increasingly expands into remote, distributed, and AI-mediated environments, fundamental challenges are emerging. Geographic separation, unequal access to technology, shifting power dynamics, and the participation of vulnerable or historically marginalized communities make it increasingly difficult to preserve the core values of participatory design, including agency, reciprocity, trust, and shared decision-making.

My current research addresses these challenges by investigating how emerging technologies can strengthen participatory design rather than replace it. Building on work in participatory design ethics, collaborative technologies, and digital co-design environments, I research ethical design methodologies, accessible collaborative systems, and AI-supported design tools that enable equitable participation across diverse communities. My long-term research vision is to implement and investigate how emerging technologies can fundamentally transform collaborative design, collective decision-making, and community participation across diverse social contexts. I envision research that combines artificial intelligence, immersive technologies, and computational design methods to create adaptive collaborative environments that empower people to collectively solve complex societal problems while preserving human agency, ethics, and inclusivity.

1 EXISTING RESEARCH CONTRIBUTIONS

1.1 Participatory design through design fiction

My research contributes both methodological frameworks and technological tools that expand how participation can occur in distributed and digitally mediated collaborative environments. One stream of my work investigates how hybrid and remote participatory design can be structured to preserve engagement, presence, and agency. In this space, I have designed and studied digital co-design environment LiveMâché[1] that allowed geographically distributed participants to collaboratively ideate and reflect. The system enabled disaster management experts to collaborate and ideate on futuristic disaster scenarios and how different technological solutions can help navigating them. This systems explore how interfaces can balance openness with facilitation, and how digital tools can scaffold but not prescribe the dynamics of participation.

1.2 Immersive and Non-Immersive Environments facilitating co-design

A second thread of my work engages with embodied and expressive modalities of collaboration. In this project avatars enabling agency and embodied presence in collaboration, I examined how avatar-mediated spaces can foster or hinder participants' sense of presence, voice, and agency in collaborative activities. I deployed Mozilla Hubs[2] for this participatory design research on to understand needs of students with dyslexia. Primary school teachers with experience of teaching students with dyslexia were participants of this research. This research revealed how even subtle representational choices in digital environments shape the interpersonal and ethical dimensions of participation insights that inform my ongoing design of systems supporting equitable co-design.

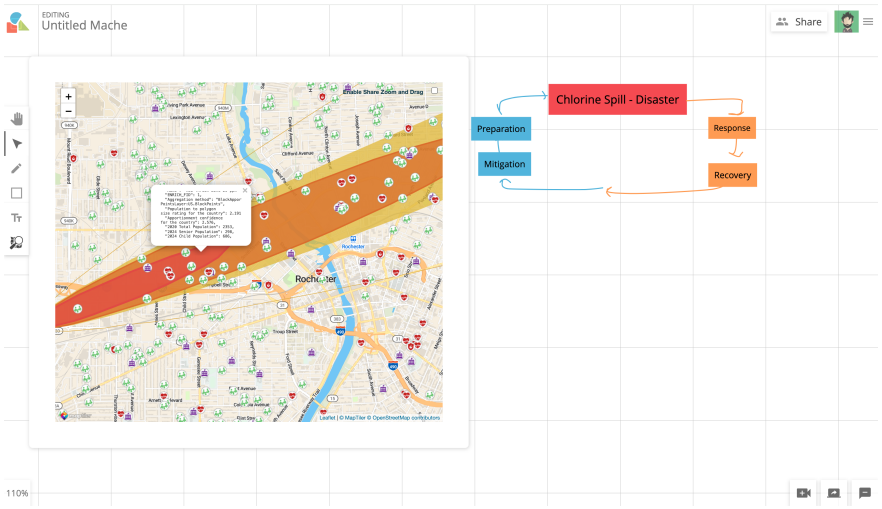


Figure 1: LiveMâché enabling participants to collaborate on a fictional disaster event scenario.

1.3 Fostering an Inclusive Community Among Electrical and Computer Engineering (ECE) Students with Mixed-Reality Technologies at a Hispanic-Serving Institution (HSI)

Our participatory design research `https://par.nsf.gov/biblio/10620946` highlighted challenges present in hybrid methods of learning, collaborating, self-representation, and socializing online for ECE students. Students felt a distinct lack of sense of community, which participants reported led to a lack of motivation and emotional support. We took a multiple-sources qualitative research approach. First, we conducted an interview based study, which is aimed at understanding the first research question. In the second study, we conducted a series of participatory design (PD) sessions with students to begin to develop designs for a mixed-reality space that addresses the community. All the students suggested incorporating avatars or physical representations of online students into the virtual classroom, and vice versa, to reflect their presence and movement accurately. We also recognized the need to develop social proximity as a feature to enhance the sense of community among students.



Figure 2: Screenshot of participants using the prototype MR system. A virtual classroom was created in Gather.town based on the classroom in which session 3 took place. Both remote and in-person students' presence is represented by virtual avatars.

2 FUTURE RESEARCH AGENDA

Across my existing projects, I combined qualitative analysis, design-based research, and prototype development to bridge theoretical reflection with practical toolmaking. My work contributes to conversations in participatory design, HCI, and design ethics by articulating new frameworks for understanding participation as situated, mediated, and relational in technologically rich contexts.

My future research builds on my work in participatory design to investigate how people form, negotiate, and transform their understanding of possible futures through interaction with others, technologies, and their surrounding social environments. I am particularly interested in developing computational and quantitative approaches that can complement participatory methods without reducing the complexity and contextual nature of human experiences. I envision three interrelated directions:

1. **AI-Augmented Participatory Design.** I plan to investigate how machine learning and generative AI can support participation without replacing human judgment or flattening differences among participants. AI systems could, for example, synthesize participant input and generate alternative representations of ideas during participatory activities. Rather than treating AI as an autonomous decision-maker, I am interested in understanding it as a mediator within a participatory system. This raises fundamental questions about how the introduction of an AI agent changes participation: Whose perspectives become more visible? How does AI-mediated synthesis influence group and individual understanding? Can AI help participants recognize perspectives that they might otherwise overlook? I would combine participatory experiments with computational analysis to examine how different forms of AI mediation influence individual and collective participation.

2. **Empathy in Distributed Participation.** A second direction examines how empathy is expressed, experienced, and enacted when participation occurs across distributed physical and digital environments. Participatory design often assumes that interaction helps participants understand one another, yet the mechanisms through which such understanding develops remain difficult to observe and measure. I am interested in treating empathy not simply as an individual psychological characteristic, but as something that can emerge through interactions among participants, artifacts, technologies, and social contexts. I plan to investigate how different forms of distributed participation influence participants' perceptions of one another and how these perceptions subsequently affect communication, trust, disagreement, and co-creation. Combining qualitative accounts of participants' experiences with quantitative measures of interaction could provide a way to examine how interpersonal processes contribute to larger patterns of collective participation.

3. **Individual and Collective Dynamics of Imagining Futures.** My third direction focuses on how individuals and groups construct, negotiate, and revise possible futures through participation. My previous work suggests that people's imagined futures are not formed in isolation. They emerge through interactions among individuals, communities, institutions, technologies, cultural expectations, and living conditions. Yet we still have limited understanding of how these interconnected factors collectively shape the futures people perceive as possible, desirable, or attainable. I am interested in developing methods that can connect these individual experiences to broader patterns of collective behavior. Participatory activities using different design methods could generate qualitative accounts of imagined futures, while computational approaches could investigate how participation structures and interaction patterns influence the emergence and transformation of ideas across individuals and groups.

My broader goal is to shape how computing systems support people in working together on problems that matter, from local communities to global challenges. Over the long term, I want to establish participation as a design principle in computer science, so that AI, collaborative platforms, and immersive environments

are built around how people learn, deliberate, and imagine futures together, not around efficiency alone. This means building systems that people can shape, studying how those systems change human judgment and trust, and developing methods that connect individual experience to collective outcomes. Across these directions, I hope to contribute to human-computer interaction, computer-supported cooperative work, and AI, and to help ensure that the technologies we build expand human agencies.